



Tutorial 7: Database Integration and Data Linkage

+ Question 1

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- We consider the following three parties related to the Olympics information system involving swimming events:
 - Local Organisation Committee (**LOC**) for an Olympiad (e.g., the London Game in 2012)
 - **Result**(EventID, CompID, Position, Time)
 - International Olympic Committee (**IOC**)
 - **Competitor**(ID, Country, Name)
 - **OlympicRecord**(EventID, CompID, Olympiad, Time)
 - **FINA**, international swimming federation
 - **Athlete**(ID, Country, Name)
 - **WorldRecord**(EventID, AthID, Year, Time)

+ Q1: Olympic Information System



- Local Organisation Committee (LOC)
- International Olympic Committee (IOC)
- International Swimming Federation (FINA)

LOC

Result
EventID
CompID
Position
Time

FINA

Athlete
ID
Country
Name

World Record
EventID
AthID
Year
Time

Competitor
ID
Country
Name

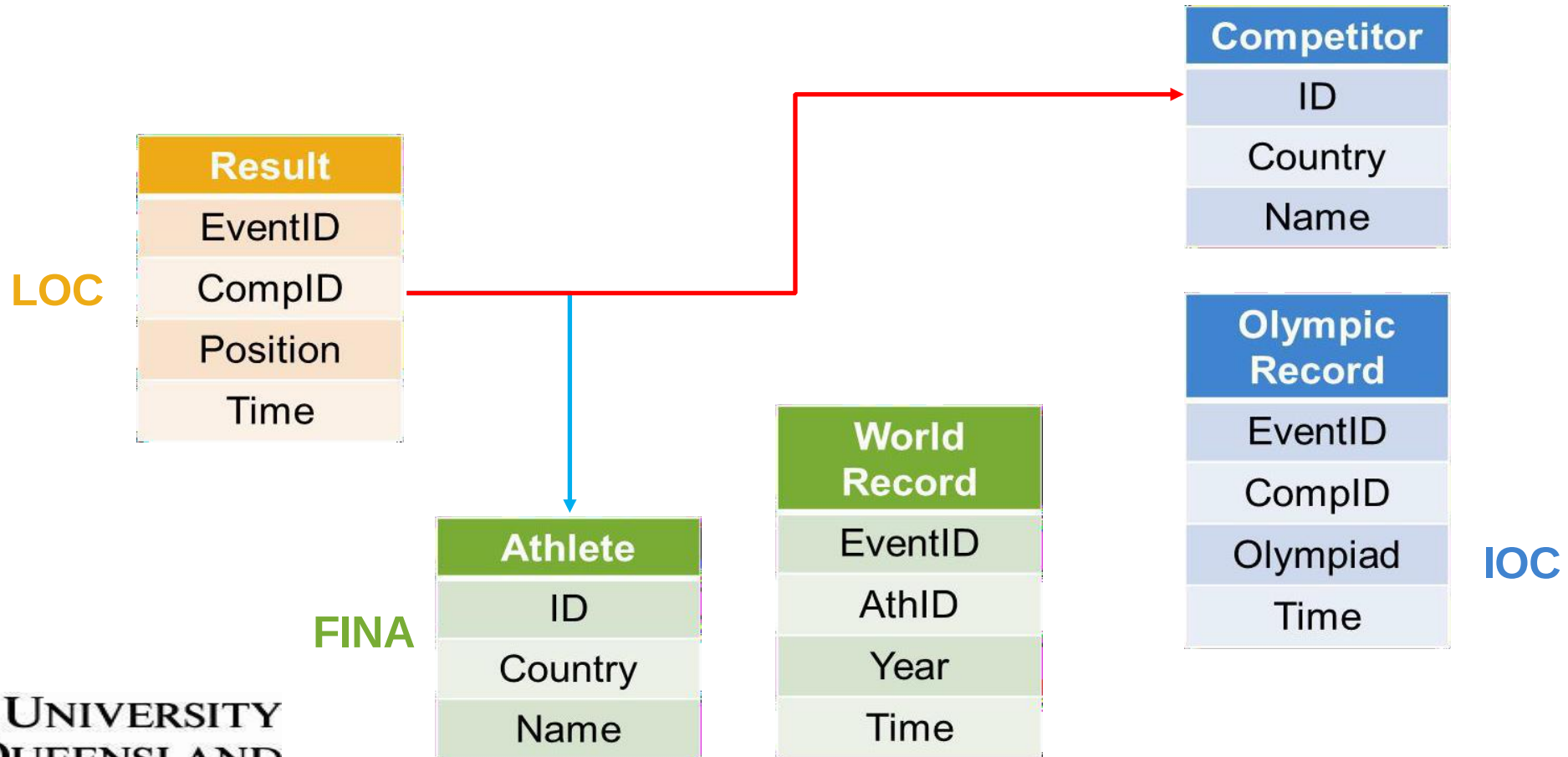
Olympic Record
EventID
CompID
Olympiad
Time

IOC

+ Q1: Olympic Information System

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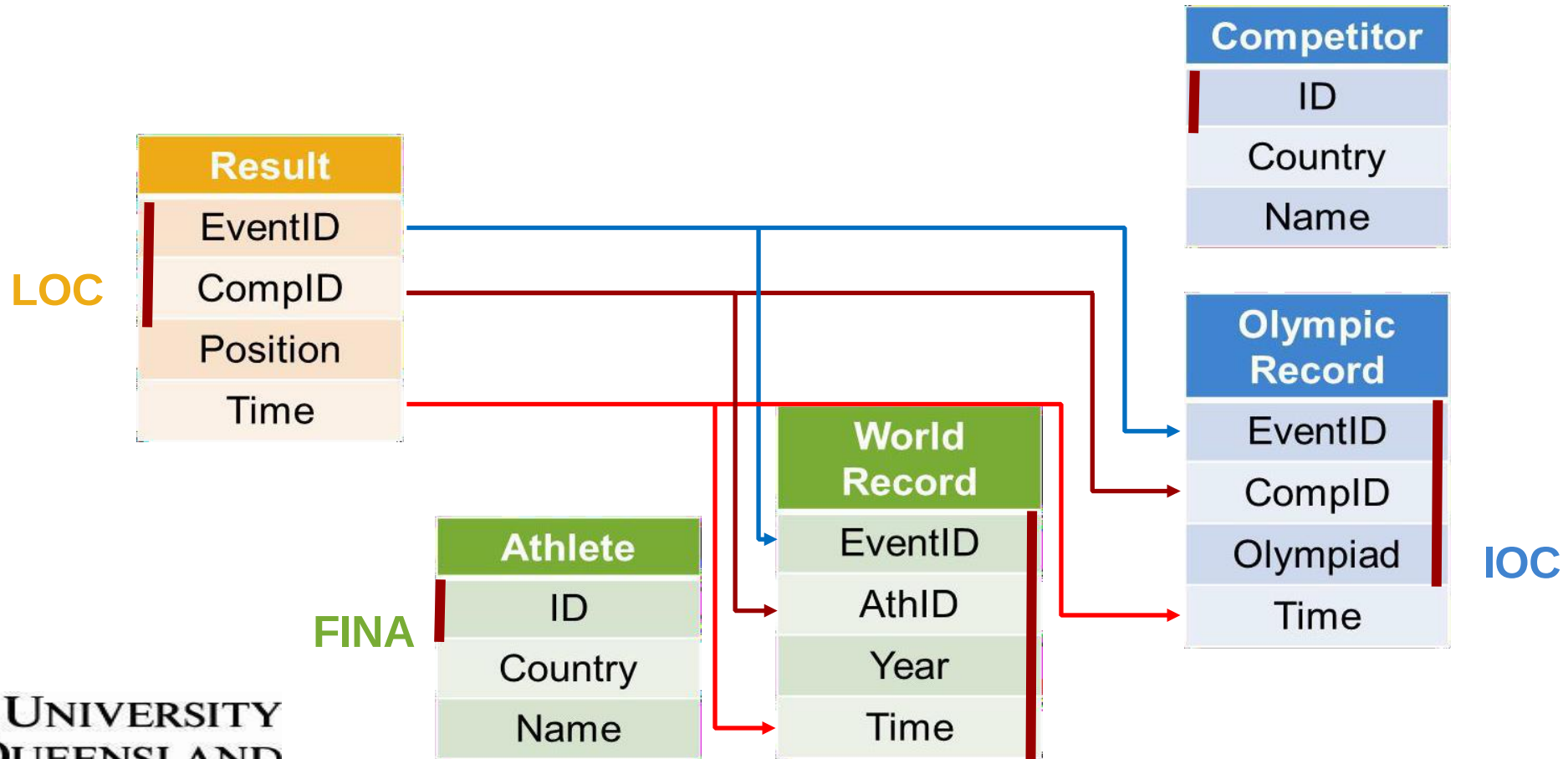
- All competitors participating the Game organised by the **LOC** are registered with the **IOC**, and also **FINA**.



+ Q1: Olympic Information System

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- Record-breaking game results are saved in both **LOC**, **IOC** and **FINA**.

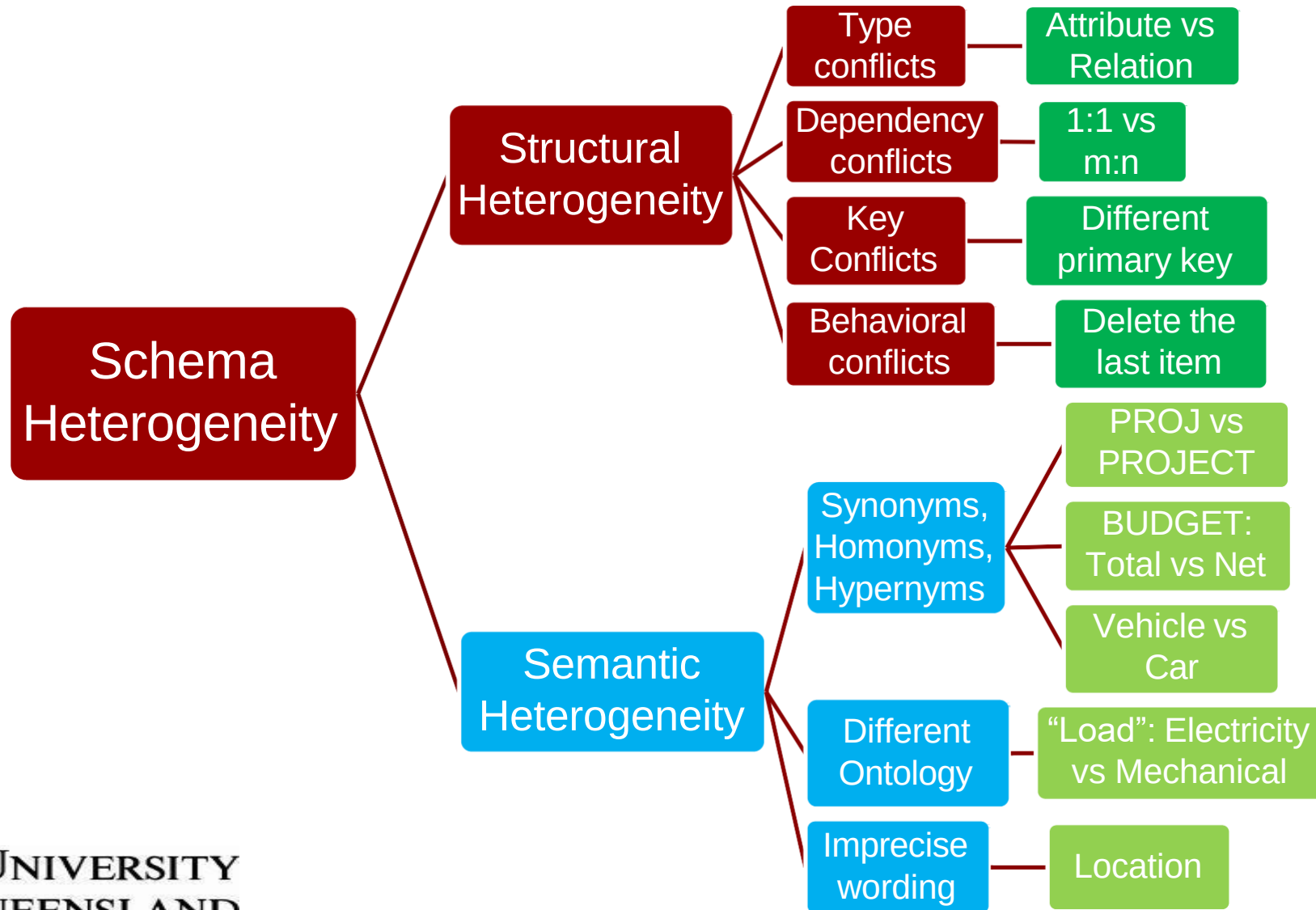


+ Q1: Olympic Information System

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- (a) Identify possible **semantic heterogeneity** when integrating these three independently developed databases into **GoldMedalist**, and discuss possible solutions.

Semantic Heterogeneity



- Example (Integration)
 - WORKER(wNumber, name, title, salary)
PROJECT(pNumber, pName, budget)
CLIENT(cName, address)
WORKS IN(wNumber, pNumber, responsibility, duration)
CONTRACTED BY(pNumber, cName, contractNo)
 - EMP(eNo, eName, title)
PROJ(pNo, pName, budget, loc, cName)
ASG(eNo, pNo, resp, dur)
PAY(title, sal)

+ Q1-a

GoldMedalist(CompID, EventID, Time)
at least 3 attributes

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- EventID: Standardised by FINA and adopted by IOC.
- FINA and IOC have been working together for many decades.
- No semantic heterogeneity.

LOC

Result
EventID
CompID
Position
Time

FINA

Athlete
ID
Country
Name

World Record
EventID
AthID
Year
Time

Competitor
ID
Country
Name

Olympic Record
EventID
CompID
Olympiad
Time

IOC

- FINA.Athlete.ID $\neq$ IOC.Competitor.ID
 - They organise ID in different ways before they start collaboration
 - Maintain a mapping in table, ask athlete from FINA to supply sporting federation ID when registering in IOC

LOC

Result
EventID
CompID
Position
Time

FINA

Athlete
ID
Country
Name

World Record
EventID
AthID
Year
Time

Competitor
ID
Country
Name

Olympic Record
EventID
CompID
Olympiad
Time

IOC

+ Q1-a

14

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LOC

Result
EventID
CompID
Position
Time

FINA

Athlete
ID
Country
Name

World
Record

World Record
EventID
AthID
Year
Time

Competitor

Competitor
ID
Country
Name
SportingFedID

Olympic
Record

Olympic Record
EventID
CompID
Olympiad
Time

IOC

+ Q1-b

- (b) Now the LOC wants to integrate these three databases into the following table that shows all swimming event Gold medalists in the Game:
 - GoldMedalist(CompID, EventID, Time, OlympicRecord, WorldRecord)

LOC

Result
EventID
CompID
Position
Time

FINA

Athlete
ID
Country
Name

World
Record

World Record
EventID
AthID
Year
Time

Olympic
Record

Olympic Record
EventID
CompID
Olympiad
Time

Competitor

ID
Country
Name

IOC

+ Q1-b

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- Now the LOC wants to integrate these three databases into the following table that shows all swimming event Gold medalists in the Game:
 - **GoldMedalist**(CompID, EventID, Time, OlympicRecord, WorldRecord)
- Use SQL to construct GoldMedallist.
 - **Result**(EventID, CompID, Position, Time)
 - **OlympicRecord**(EventID, CompID, Olympiad, Time)
 - **WorldRecord**(EventID, AthID, Year, Time)

+ Q1-b

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CREATE VIEW GoldMedalist

(CompID, EventID, Time, OlympicRecord, WorldRecord) **AS**

SELECT B.CompID, B.EventID, B.Time, IR.Time, WR.Time

FROM Result B, OlympicRecord IR, WorldRecord WR

WHERE

IR.EventID = B.EventID **AND**

WR.EventID = B.EventID **AND**

B.Position = 1;

- (c) Assume the ABC Television wants to create the following table to show all the swimming records set at the Game organised by the LOC:

- **NewRecord**(EventID, CompID, Record, Time)

where Record is either "World" or "Olympic". Show an SQL query computing NewRecord.

- Solution

- **WorldRecord** <= **OlympicRecord**
 - Break the WorldRecord
 - **Time** < **WorldRecord**
 - Break the OlympicRecord
 - **Time** >= **WorldRecord** and **Time** < **OlympicRecord**

+ Q1-c

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```
CREATE VIEW NewRecord(EventID, CompID, Record, Time) AS
  SELECT EventID, CompID, "World", Time
  FROM GoldMedallist G
  WHERE Time < G.WorldRecord
UNION
  SELECT EventID, CompID, "Olympic", Time
  FROM GoldMedallist G
  WHERE Time >= G.WorldRecord AND Time < G.OlympicRecord
```


+ Q2. Edit Distance

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- The **edit distance** between two strings is the **minimum** number of operations to transform one string to another
 - Operations:
delete, insert or substitution/replace one character
 - Normalization to $[0, 1]$
 - Divided by $\max(|A|, |B|)$
- What's the edit distance?
 - 'John', 'Jon' $1/4$
 - 'John', 'Josh' $2/4$
 - 'Smith', 'Sitch' $2/5$

+ Q2. Jaccard Distance

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- Jaccard Distance
 - Based on string tokenization
 - Breaking a stream of **text** up into words, phrases, symbols, or other forms of elements called tokens
 - An **n-gram** is a contiguous sequence of items from given sequence of text or speech

1-gram sequence

It was the best of times

I, t, #, w, a, s, #, t, h, e, #, b, e, s, t, #, o, f, #, t, i, m, e, s

2 gram sequence

It, t#, #w, wa, as, s#, #t, th, he, e#, #b, be, es, st, t#, #o, of, f#, #t, ti, im, me, es

3 gram sequence

It#, t#w, #wa, was, as#, s#t, #th, the, he#, e#b, #be, bes, est, st#, t#o, #of, of#, f#t, #ti, tim, ime, mes



+ Q2. Jaccard Distance

25

■ Jaccard coefficient

$$J(A, B) = \frac{|\text{intersect}(A, B)|}{|\text{union}(A, B)|} = \frac{|A \cap B|}{|A \cup B|}$$

$$= \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

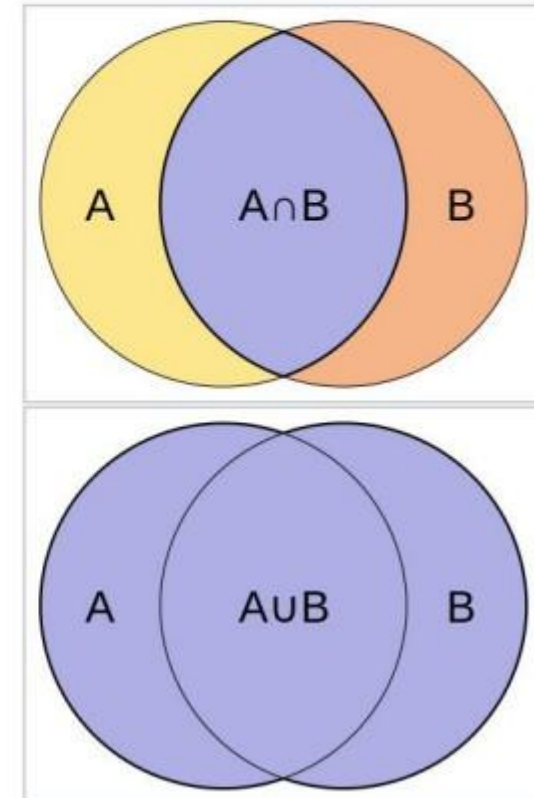
$$0 \leq J(A, B) \leq 1$$

■ Jaccard distance

$$d_J(A, B) = 1 - J(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}$$

$$= \frac{|A \cup B| - |A \cap B|}{|A \cup B|}$$

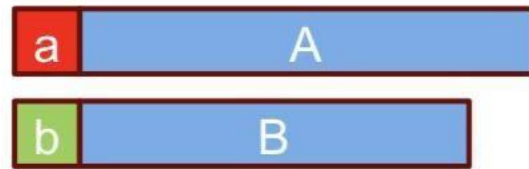
dissimilarity



+ Q2. Edit Distance Computation

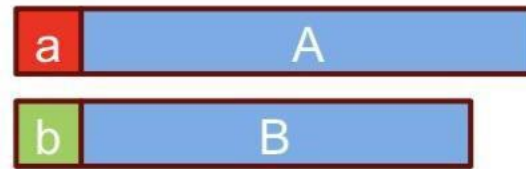
26

■ Intuition

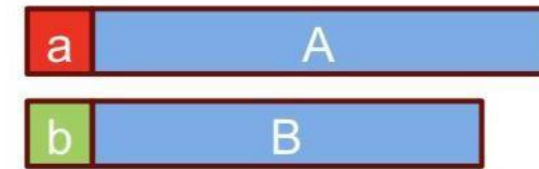


$$d(A, B) + 1$$

replace



$$d(A, "b" + B) + 1$$



$$d("a" + A, B) + 1$$

$$A_0 = a_1 a_2 a_3 a_4 \cdots a_{m-1} a_m$$
$$B_0 = b_1 b_2 b_3 b_4 \cdots b_{n-1} b_n$$



$$A_0 = \mathbf{a_1} a_2 a_3 a_4 \cdots a_{m-1} a_m$$
$$B_0 = \mathbf{b_1} b_2 b_3 b_4 \cdots b_{n-1} b_n$$



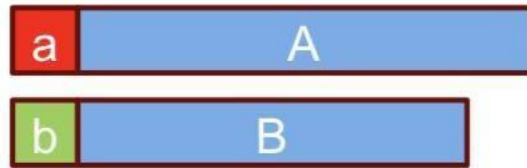
$$d(A_0, B_0)$$
$$= d(A_1, B_1) + \text{Cost}_{\text{replace}}$$

$$A_1 = \mathbf{a_1} a_2 a_3 a_4 \cdots a_{m-1} a_m$$
$$B_1 = \mathbf{a_1} b_2 b_3 b_4 \cdots b_{n-1} b_n$$

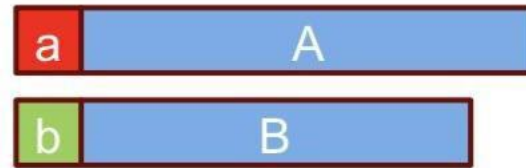
+ Q2. Edit Distance Computation

27

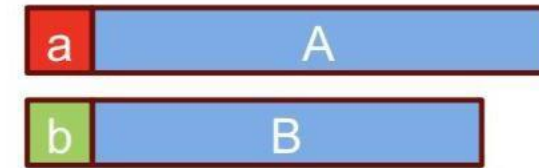
■ Intuition



$$d(A, B) + 1$$



$$d(A, "b" + B) + 1$$



$$d("a" + A, B) + 1$$

delete

$$A_0 = a_1 a_2 a_3 a_4 \cdots a_{m-1} a_m$$

$$B_0 = b_1 b_2 b_3 b_4 \cdots b_{n-1} b_n$$



$$A_0 = \mathbf{a}_1 a_2 a_3 a_4 \cdots a_{m-1} a_m$$

$$B_0 = b_1 b_2 b_3 b_4 \cdots b_{n-1} b_n$$



$$A_1 = a_2 a_3 a_4 \cdots a_{m-1} a_m$$

$$B_0 = b_1 b_2 b_3 b_4 \cdots b_{n-1} b_n$$

$$d(A_0, B_0)$$

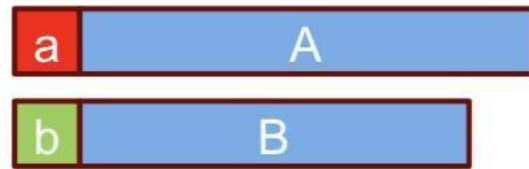
$$= d(A_1, B_0) + \text{Cost}_{\text{delete}}$$

$$= d(A_1, b_1 + B_1) + \text{Cost}_{\text{delete}}$$

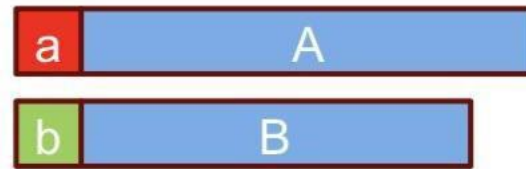
+ Q2. Edit Distance Computation

28

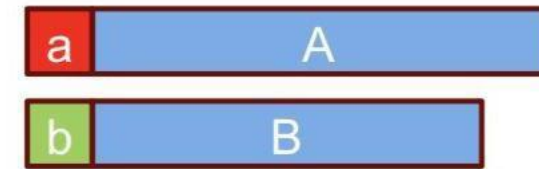
■ Intuition



$$d(A, B) + 1$$



$$d(A, "b" + B) + 1$$



$$d("a" + A, B) + 1$$

insert

$$A_0 = a_1 a_2 a_3 a_4 \cdots a_{m-1} a_m$$

$$B_0 = b_1 b_2 b_3 b_4 \cdots b_{n-1} b_n$$

$$A_1 = \mathbf{a_0} a_1 a_2 a_3 a_4 \cdots a_{m-1} a_m$$

$$B_0 = b_1 b_2 b_3 b_4 \cdots b_{n-1} b_n$$

$$d(A_0, B_0) = d(A_1, B_0) + \text{Cost}_{\text{insert}}$$

$$= d(a_0 + A_0, B_0) + \text{Cost}_{\text{insert}}$$

+ Q2. Edit Distance Computation

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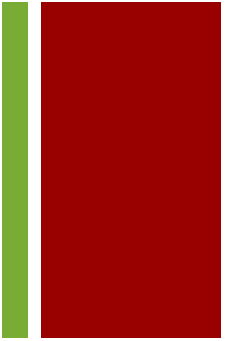
$$d_{ij} = \begin{cases} d_{i-1,j-1} & \text{for } a_j = b_i \\ \min \begin{cases} d_{i-1,j} + w_{\text{del}}(b_i) \\ d_{i,j-1} + w_{\text{ins}}(a_j) \\ d_{i-1,j-1} + w_{\text{sub}}(a_j, b_i) \end{cases} & \text{for } a_j \neq b_i \end{cases}$$

University Queensland

Queensland University



		U	N	I	V	E	R	S	I	T	Y	#	Q	U	E	E	N	S	L	A	N	D
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$$d_{ij} = \begin{cases} d_{i-1, j-1} & (a_j = b_i) \\ \min \begin{pmatrix} d_{i-1, j+1} \\ d_{i, j-1} + 1 \\ d_{i-1, j-1} + 1 \end{pmatrix} & (a_j \neq b_i) \end{cases}$$

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		U	N	I	V	E	R	S	I	T	Y	#	Q	U	E	E	N	S	L	A	N	D
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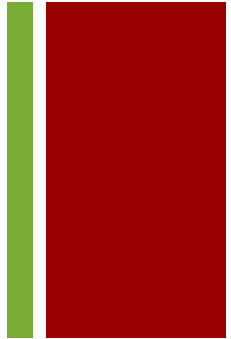
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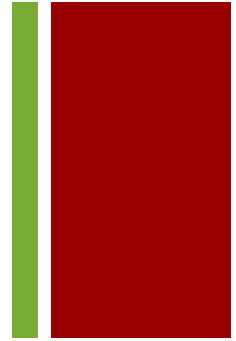
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		U	N	I	V	E	R	S	I	T	Y	#	Q	U	E	E	N	S	L	A	N	D
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
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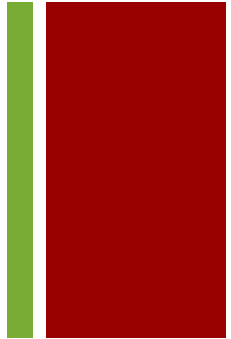
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E	4	3	3	3	4	4	5	6	7	8	9	10	11	12	12	11	12	13	14	15	16	17
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Y	21	20	19	18	17																	



$$d_{ij} = \begin{cases} d_{i-1, j-1} & (a_j = b_i) \\ \min \begin{pmatrix} d_{i-1, j} + 1 \\ d_{i, j-1} + 1 \\ d_{i-1, j-1} + 1 \end{pmatrix} & (a_j \neq b_i) \end{cases}$$



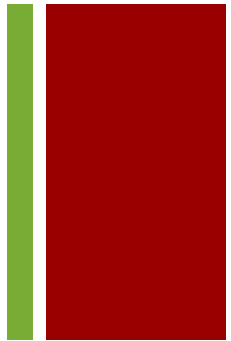
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Q	1	1	2	3	4	5	6	7	8	9	10	11	11	12	13	14	15	16	17	18	19	20
U	2	1	2	3	4	5	6	7	8	9	10	11	12	11	12	13	14	15	16	17	18	19
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N	5	4	3	4	4	5	5	6	7	8	9	10	11	12	13	12	11	12	13	14	15	16
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I	14	13	12	11	12	13																
V	15	14	13	12	11	12																
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$$d_{ij} = \begin{cases} d_{i-1, j-1} & (a_j = b_i) \\ \min \begin{pmatrix} d_{i-1, j+1} \\ d_{i, j-1} + 1 \\ d_{i-1, j-1} + 1 \end{pmatrix} & (a_j \neq b_i) \end{cases}$$

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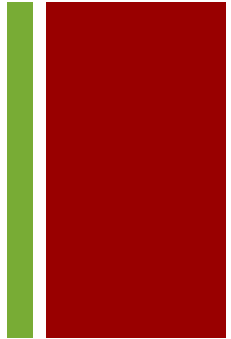
		U	N	I	V	E	R	S	I	T	Y	#	Q	U	E	E	N	S	L	A	N	D
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Q	1	1	2	3	4	5	6	7	8	9	10	11	11	12	13	14	15	16	17	18	19	20
U	2	1	2	3	4	5	6	7	8	9	10	11	12	11	12	13	14	15	16	17	18	19
E	3	2	2	3	4	4	5	6	7	8	9	10	11	12	11	12	13	14	15	16	17	18
E	4	3	3	3	4	4	5	6	7	8	9	10	11	12	12	11	12	13	14	15	16	17
N	5	4	3	4	4	5	5	6	7	8	9	10	11	12	13	12	11	12	13	14	15	16
S	6	5	4	5	5	5	6	5	6	7	8	9	10	11	12	13	12	11	12	13	14	15
L	7	6	5	6	6	6	6															
A	8	7	6	7	7	7	7															
N	9	8	7	8	8	8	8															
D	10	9	8	9	9	9	9															
#	11	10	9	10	10	10	10															
U	12	11	10	11	11	11	11															
N	13	12	11	12	12	12	12															
I	14	13	12	11	12	13	13															
V	15	14	13	12	11	12	13															
E	16	15	14	13	12	11	12															
R	17	16	15	14	13	12	11															
S	18	17	16	15	14	13	12															
I	19	18	17	16	15	14	13															
T	20	19	18	17	16	15	14															
Y	21	20	19	18	17	16	15															



$$d_{ij} = \begin{cases} d_{i-1, j-1} & (a_j = b_i) \\ \min \begin{pmatrix} d_{i-1, j} + 1 \\ d_{i, j-1} + 1 \\ d_{i-1, j-1} + 1 \end{pmatrix} & (a_j \neq b_i) \end{cases}$$

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		U	N	I	V	E	R	S	I	T	Y	#	Q	U	E	E	N	S	L	A	N	D
	0	1	2	3	4	5	6	7	8	9	10	$d_{ij} = \begin{cases} d_{i-1, j-1} & (a_j = b_i) \\ \min \begin{pmatrix} d_{i-1, j} + 1 \\ d_{i, j-1} + 1 \\ d_{i-1, j-1} + 1 \end{pmatrix} & (a_j \neq b_i) \end{cases}$										
Q	1	1	2	3	4	5	6	7	8	9	10											
U	2	1	2	3	4	5	6	7	8	9	10											
E	3	2	2	3	4	4	5	6	7	8	9											
E	4	3	3	3	4	4	5	6	7	8	9											
N	5	4	3	4	4	5	5	6	7	8	9	10	11	12	13	12	11	12	13	14	15	16
S	6	5	4	5	5	5	6	5	6	7	8	9	10	11	12	13	12	11	12	13	14	15
L	7	6	5	6	6	6	6	6	6	7	8	9	10	11	12	13	13	12	11	12	13	14
A	8	7	6	7	7	7	7	7	7	7	8	9	10	11	12	13	14	13	12	11	12	13
N	9	8	7	8	8	8	8	8	8	8	8	9	10	11	12	13	13	14	13	12	11	12
D	10	9	8	9	9	9	9	9	9	9	9	9	10	11	12	13	14	14	14	13	12	11
#	11	10	9	10	10	10	10	10	10	10	10	9	10	11	12	13	14	15	15	14	13	12
U	12	11	10	11	11	11	11	11	11	11	11	10										
N	13	12	11	12	12	12	12	12	12	12	12	11										
I	14	13	12	11	12	13	13	13	12	13	13	12										
V	15	14	13	12	11	12	13	14	13	13	14	13										
E	16	15	14	13	12	11	12	13	14	14	14	14										
R	17	16	15	14	13	12	11	12	13	14	15	15										
S	18	17	16	15	14	13	12	11	12	13	14	15										
I	19	18	17	16	15	14	13	12	11	12	13	14										
T	20	19	18	17	16	15	14	13	12	11	12	13										
Y	21	20	19	18	17	16	15	14	13	12	11	12										



+

		U	N	I	V	E	R	S	I	T	Y	#	Q	U	E	E	N	S	L	A	N	D	
	0	1	2	3	4	5	6	7	8	9	10	$d_{ij} = \begin{cases} d_{i-1, j-1} & (a_j = b_i) \\ \min \begin{pmatrix} d_{i-1, j+1} \\ d_{i, j-1} + 1 \\ d_{i-1, j-1} + 1 \end{pmatrix} & (a_j \neq b_i) \end{cases}$											
Q	1	1	2	3	4	5	6	7	8	9	10												
U	2	1	2	3	4	5	6	7	8	9	10												
E	3	2	2	3	4	4	5	6	7	8	9												
E	4	3	3	3	4	4	5	6	7	8	9												
N	5	4	3	4	4	5	5	6	7	8	9	10	11	12	13	12	11	12	13	14	15	16	
S	6	5	4	5	5	5	6	5	6	7	8	9	10	11	12	13	12	11	12	13	14	15	
L	7	6	5	6	6	6	6	6	6	7	8	9	10	11	12	13	13	12	11	12	13	14	
A	8	7	6	7	7	7	7	7	7	7	8	9	10	11	12	13	14	13	12	11	12	13	
N	9	8	7	8	8	8	8	8	8	8	8	9	10	11	12	13	13	14	13	12	11	12	
D	10	9	8	9	9	9	9	9	9	9	9	9	10	11	12	13	14	14	14	13	12	11	
#	11	10	9	10	10	10	10	10	10	10	10	9	10	11	12	13	14	15	15	14	13	12	
U	12	11	10	11	11	11	11	11	11	11	11	10	10	10	11	12	13	14	15	15	14	13	
N	13	12	11	12	12	12	12	12	12	12	12	11	11	11	11	12	12	13	14	15	15	14	
I	14	13	12	11	12	13	13	13	12	13	13	12	12	12	12	12	13	13	14	15	16	15	
V	15	14	13	12	11	12	13	14	13	13	14	13	13	13	13	13	13	14	14	15	16	16	
E	16	15	14	13	12	11	12	13	14	14	14	14	14	14	14	13	13	14	14	15	15	16	17
R	17	16	15	14	13	12	11	12	13	14	15	15	15	15	14	14	14	15	15	16	16	17	
S	18	17	16	15	14	13	12	11	12	13	14	15	16	16	15	15	15	14	15	16	17	17	
I	19	18	17	16	15	14	13	12	11	12	13	14	15	16	16	16	16	15	15	16	17	18	
T	20	19	18	17	16	15	14	13	12	11	12	13	14	15	16	17	17	16	16	17	17	18	
Y	21	20	19	18	17	16	15	14	13	12	11	12	13	14	15	16	17	17	17	17	18	18	

+

Q2. Edit Distance Computation

39

UNIVERSITY#QUEENSLAND

Length = 21

UNIVERSITY#QUEENSLAND

UNIVERSITY#QUNERSITYD

UNIVERSITY#UNIVERSITY

QUNIVERSITYD#UNIVERSITY

QUEIVENSLAND#UNIVERSITY

QUEIVENSLAND#UNIVERSITY

QUEENSLAND#UNIVERSITY

+ Q2. Edit Distance Computation

40

- distance = 18

*at least 18 operations should be performed
to transform string A to B*

- similarity

$$\text{sim}(a, b) = 1 - \frac{ED(a, b)}{\max(|a|, |b|)}$$

$$= 1 - \frac{18}{21} = 0.143$$

+ Q2. Jaccard Coefficient Comp.

41

■ 3-grams

A = University Queensland

Uni, niv, ive, ver, ers, rsi, sit, ity,
ty#, y#Q, #Qu, Que, uee, een, ens, nsl,
sla, lan, and

B = Queensland University

Que, uee, een, ens, nsl, sla, lan, and,
nd#, d#U, #Un, Uni, niv, ive, ver, ers,
rsi, sit, ity

+ Q2. Jaccard Coefficient Comp.

42

A Uni, niv, ive, ver, ers, rsi, sit, ity,
ty#, y#Q, #Qu, Que, uee, een, ens, nsl,
sla, lan, and

B Que, uee, een, ens, nsl, sla, lan, and,
nd#, d#U, #Un, Uni, niv, ive, ver, ers,
rsi, sit, ity

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{16}{16 + 3 + 3} = 0.727$$

$$d_J(A, B) = 1 - J(A, B) = 0.273$$

+ Q2–(d) Conclusion

43

- similarity

- Edit Distance: 0.143
- Jaccard Coefficient: 0.727

- dissimilarity

- Edit Distance: $1 - 0.143 = 0.857$
- Jaccard Distance: $1 - 0.727 = 0.273$
- Jaccard distance is less sensitive to word orders

University Queensland

Queensland University

+ Q3 (a)

43

- Answer the following questions:
- (a) What problem is solved by Cosine Similarity using the TF/IDF measure?
- **Answer:** The Cosine Similarity using TF/IDF is computed between two groups of terms (or words). So, the granularity of this computation is based on the sets of terms (words) that belong to different documents (or different groups of records) in order to find out their overall similarity.

+ Q3 (b)

43

- Answer the following questions:
- (b) If the numerical numbers of 1,000 and 999 are regarded as similar data items, how do we compute this kind of similarity?
- Answer: This is called “Numeric Similarity”, two numbers are regarded as similar if they are in a range.

See the following two tables' schema:

DRIVER

Driver-License#	First-Name	Middle-Name	Family-Name	DoB	Gender	Address	Home-Phone-Number	Expiry-Date
-----------------	------------	-------------	-------------	-----	--------	---------	-------------------	-------------

TAXPAYER

Tax-File #	First-Name	Surname	Resident Status	DoB	Sex	Post-Code	eMail	Contact#
------------	------------	---------	-----------------	-----	-----	-----------	-------	----------

- We want to create a mapping table namely, PERSON based on these two given
- tables:

PERSON

Tax-File #	Driver-License#
------------	-----------------

How do we do it?

- This is a problem of Record Matching discussed in Week05 Lecture Notes, #66 - #68.
- In the process, we may also prioritize the attributes (i.e., assigning the weights on the certain attributes of the schema) to emphasize the importance of an individual attribute. Then the similarities between the pairs of relevant attributes can be calculated.

- For example, the weight (i.e., the importance) of attributes matching of “Driver.Family-Name” with “Taxpayer.Surname” can be assigned an overall weight of 25%, “Driver.First-Name” with “Taxpayer.First-Name” can be assigned an overall weight of 15%, “Driver.DoB” with “Taxpayer.DoB” can be assigned an overall weight of 25%, “Driver.Gender” with “Taxpayer.Sex” can be assigned an overall weight of 25%, “Driver.address” with “Taxpayer.Post-Code” can be assigned an overall weight of 5%, “Driver.Home-Phone#” with “Taxpayer.Contact#” are assigned an overall weight of 5%.
- Also, when we calculate the similarity between attributes, we can decide a threshold to qualify the similarity before the assigned weight can be applied for the result of this matching calculation of attributes.

- See the following table of this idea:

Input Weight	Driver	Taxpayer	Threshold of Similarity
25%	Family-Name	Surname	100%
15%	First-Name	First-Name	70%
25%	DoB	DoB	100%
25%	Gender	Sex	100%
5%	Address	Post-Code	40%
5%	Home-Phone number	Contact#	100%

- The overall threshold for computing on an output entry can be decide if $\text{Weights} \geq 90\%$.
- The resulting output will be the combination of the selected output attributes from each of the table as (Driver.License# Taxpayer.Tax-File#):

Tax-File #	Driver-License#
------------	-----------------

- We may also include an extra attributed for the credibility/probability of this matching computation as:

Tax-File #	Driver-License#	Probability (%)
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